

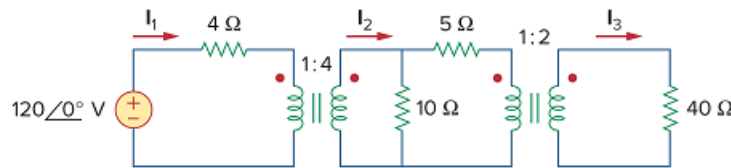
Problem

Refer to the circuit in Fig. 13.125 on the following page.

(a) Find currents I_1 , I_2 , and I_3 .

(b) Find the power dissipated in the $40\text{-}\Omega$ resistor.

Figure 13.125



Step-by-step solution

Step 1 of 10

Refer to Figure 13.125 in the textbook.

(a)

The general rule for eliminating the transformer and reflecting the secondary circuit to the primary side is divide the secondary impedance by n^2 .

First move the right most secondary resistance to primary (middle).

Calculate the modified resistance.

$$R_{40\Omega} = \frac{40}{n^2}$$

Here,

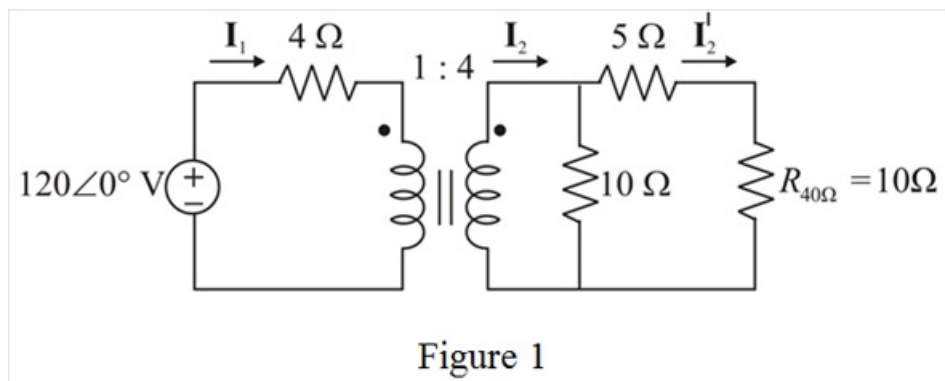
$$\begin{aligned} n &= \frac{N_2}{N_1} \\ &= \frac{2}{1} \\ &= 2 \end{aligned}$$

Substitute 2 for n .

$$\begin{aligned} R_{40\Omega} &= \frac{40}{(2)^2} \\ &= \frac{40}{4} \\ &= 10\Omega \end{aligned}$$

Step 2 of 10

Draw the modified circuit diagram.

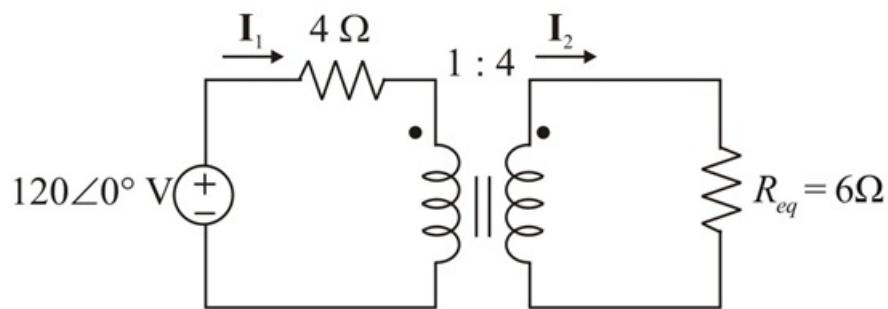


Step 3 of 10

Calculate the equivalent resistance from the secondary side of Figure 1.

$$\begin{aligned} R_{eq} &= 10 \parallel (5 + 10) \\ &= 10 \parallel 15 \\ &= \frac{15 \times 10}{15 + 10} \\ &= 6 \, \Omega \end{aligned}$$

Draw the modified circuit diagram.



Step 4 of 10

Now move the secondary resistance to primary.

Calculate the modified resistance.

$$R_{eq1} = \frac{6}{n^2}$$

Here,

$$\begin{aligned} n &= \frac{N_2}{N_1} \\ &= \frac{4}{1} \\ &= 4 \end{aligned}$$

Substitute 4 for n .

$$\begin{aligned} R_{eq1} &= \frac{6}{(4)^2} \\ &= \frac{6}{16} \\ &= 0.375 \, \Omega \end{aligned}$$

Draw the modified circuit diagram.

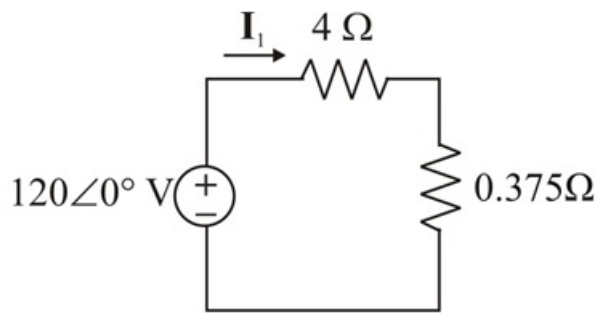


Figure 3

Apply Ohm's law to calculate the current, I_1 .

$$\begin{aligned} I_1 &= \frac{120\angle 0^\circ}{4 + 0.375} \\ &= \frac{120}{4.375} \\ &= 27.428\angle 0^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current, I_1 is $27.428\angle 0^\circ \text{ A}$.

Calculate the value of current, I_2 from Figure 2.

$$\begin{aligned} \frac{I_2}{I_1} &= \frac{N_1}{N_2} \\ \frac{I_2}{I_1} &= \frac{1}{4} \\ I_2 &= \frac{I_1}{4} \end{aligned}$$

Substitute $27.428\angle 0^\circ \text{ A}$ for I_1 .

$$\begin{aligned} I_2 &= \frac{27.428\angle 0^\circ \text{ A}}{4} \\ &= 6.857\angle 0^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current, I_2 is $6.857\angle 0^\circ \text{ A}$.

Apply current division rule to Figure 1 to calculate primary current to right most windings.

$$\begin{aligned} I'_2 &= \left[\frac{10}{10 + (5 + 10)} \right] I_2 \\ &= \left[\frac{10}{25} \right] (6.857\angle 0^\circ \text{ A}) \\ &= 2.7428\angle 0^\circ \text{ A} \end{aligned}$$

Calculate the value of current, $\mathbf{I}_3$ from the actual circuit.

$$\frac{\mathbf{I}_3}{\mathbf{I}_2'} = \frac{N_1}{N_2}$$

$$\frac{\mathbf{I}_3}{\mathbf{I}_2'} = \frac{1}{2}$$

$$\mathbf{I}_3 = \frac{\mathbf{I}_2'}{2}$$

Substitute $2.7428\angle 0^\circ \text{ A}$ for $\mathbf{I}_2'$.

$$\begin{aligned}\mathbf{I}_3 &= \frac{2.7428\angle 0^\circ \text{ A}}{2} \\ &= 1.3714\angle 0^\circ \text{ A}\end{aligned}$$

Therefore, the value of the current, $\mathbf{I}_3$ is $\boxed{1.3714\angle 0^\circ \text{ A}}$.

Step 10 of 10

(b)

Calculate the power dissipated in the 40Ω resistor.

$$P_{40\Omega} = \frac{1}{2} \times |\mathbf{I}_3|^2 \times 40$$

Substitute 1.3714 for $|\mathbf{I}_3|$.

$$\begin{aligned}P_{40\Omega} &= \frac{1}{2} \times |1.3714|^2 \times 40 \\ &= 37.6147 \text{ W}\end{aligned}$$

Therefore, the power dissipated in the 40Ω resistor is $\boxed{37.6147 \text{ W}}$.